

Assignment 6

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Preliminaries

Package Loading

```
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 3.2.5
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 3.2.5
```

```
library(stringr)  
library(dplyr, warn.conflicts = F)
```

```
## Warning: package 'dplyr' was built under R version 3.2.5
```

```
library(purrr, warn.conflicts = F)
```

```
## Warning: package 'purrr' was built under R version 3.2.5
```

```
library(readr)
```

```
## Warning: package 'readr' was built under R version 3.2.5
```

```
library(epitools)  
library(scatterplot3d)
```

```
## Warning: package 'scatterplot3d' was built under R version 3.2.5
```

Assignment Part I

1) *Answer to Problem 1*

- a) Only F occurs: $F \cap (E \cup G)^c$
- b) both E and F but not G occur: $(F \cap E) \cap G^c$
- c) at least one event occurs: $E \cup F \cup G$

d) at least one event occurs: $(E \cap F) \cup (F \cap G) \cup (E \cap G)$

e) all three events occur: $E \cap F \cap G$

f) none of them occurs: $E^c \cap F^c \cap G^c$

g) at most one occurs: $(E \cap F^c \cap G^c) \cup (E^c \cap F^c \cap G) \cup (E^c \cap F \cap G^c) \cup (E^c \cap G^c \cap F^c)$

or: $\{(E \cap F) \cup (F \cap G) \cup (E \cap G)\}^c$

h) at most two occur: $(E \cap F \cap G)^c$

2) *Answer to Problem 2*

$$p = \frac{\binom{95}{4}}{\binom{100}{4}} = 0.812$$

```
x<-choose(95,4)
y<-choose(100,4)
x/y
```

```
## [1] 0.8118751
```

3) *Answer to Problem 3*

event A: Alice does not find her paper in drawer i

event B: the paper is in drawer j

p_i : the probability that the paper is in drawer i

d_i : her search is successful with p_i

So we have: $P(A) = 1 - p_i d_i$

If $j \neq i$, $A \cap B = B$, thus $P(A \cap B) = P(B)$, and we have

$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{P(B)}{P(A)} = \frac{p_j}{1 - p_i d_i}$$

if $j = i$, we have

$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{P(B)P(A | B)}{P(A)} = \frac{p_i(1 - d_i)}{1 - p_i d_i}$$

4) *Answer to Problem 4*

a)

i) under following situations, Boris can win: “win-win”, “win-lose-win”, “lose-win-win”, so the probability is:

$$p(\text{Boris wins}) = p_w \times p_w + (1 - p_w) \times p_w \times p_w + p_w \times (1 - p_w) \times p_w = p_w^2 + 2(1 - p_w) \times p_w^2$$

ii) under following situations, Boris can win: “draw-draw-win”, so the probability is:

$$p(\text{Boriswins}) = p_d^2 \times p_w$$

iii) under following situations, Boris can win: “win(bold)-lose(timid)-win(bold)”, “win(bold)-draw(timid)”, and “lose(bold)-win(bold)-win(bold)” so the probability is:

$$p(\text{Boriswins}) = p_w \times p_d + p_w(1 - p_d)p_w + (1 - p_w)p_w^2$$

b)

If $p_w < \frac{1}{2}$, Boris has a greater probability of losing rather than winning any one game, regardless of the type of play he uses. Despite this, the probability of winning that match with strategy (iii) can be greater than $\frac{1}{2}$, provided that p_w is close enough to $\frac{1}{2}$ and p_d is close enough to 1. For example, if $p_w = 0.45$ and $p_d = 0.9$, with strategy (iii), we have

$$p(\text{Boriswins}) = p_w \times p_d + p_w(1 - p_d)p_w + (1 - p_w)p_w^2 \approx 0.45$$

With strategies (i) and (ii), the corresponding probabilities of a win can be calculated to be approximately 0.43 and 0.36, respectively. What is happening here is that with strategy (iii), Boris is allowed to select a playing style after seeing the result of the first game, while his opponent is not. Thus, by being able to dictate the playing style in each game after receiving partial information about the match's outcome, Boris gains an advantage.

5) **Answer to Problem 5**

event A: the professor teaches her class

event B: the weather is bad

$$P(A) = P(A | B)P(B) + P(A | B^c)P(B^c)$$

$$P(A | B) = \sum_{i=k}^n \binom{n}{i} p_b^i (1 - p_b)^{n-i}$$

$$P(A | B^c) = \sum_{i=k}^n \binom{n}{i} p_g^i (1 - p_g)^{n-i}$$

$$P(A) = P(B)P(A | B) + (1 - P(B))P(A | B^c) = \sum_{i=k}^n \binom{n}{i} p_b^i (1 - p_b)^{n-i} + \sum_{i=k}^n \binom{n}{i} p_g^i (1 - p_g)^{n-i}$$

6) **Answer to Problem 6**

$$\text{a)} \binom{8}{4} \binom{10}{3} = 8400$$

```
x<-choose(8,4)
y<-choose(10,3)
x*y
```

```
## [1] 8400
```

b) There are four types of combinations of the curriculum according to higher level courses

i) all of the three higher courses are from $\{H_1, \dots, H_5\}$

the number of combinations is $\binom{7}{3}\binom{5}{3}$

ii) all of the three higher courses are from $\{H_6, \dots, H_{10}\}$ the number of combinations is $\binom{6}{2}\binom{5}{3}$

iii) one of the higher courses are from $\{H_1, \dots, H_5\}$ while two of them are from $\{H_6, \dots, H_{10}\}$

the number of combinations is $\binom{5}{1}\binom{5}{1}\binom{5}{2}$

iv) two of the higher courses are from $\{H_1, \dots, H_5\}$ while one of them are from $\{H_6, \dots, H_{10}\}$

the number of combinations is $\binom{5}{1}\binom{5}{2}\binom{5}{1}$

```
a<-choose(7,3)
b<-choose(5,3)
c<-choose(5,2)
e<-choose(6,2)
a*b+e*b+5*5*c+5*c*5
```

```
## [1] 1000
```

This is the result from the problem solutions to the textbook

```
d<-choose(10,3)
b*c+b*b+2*b*c+5*d
```

```
## [1] 1000
```

7) *Answer to Problem 7*

(a) I use two method to simulate, the results are presented below

```
deck<-rep(1:13,4)
aces<-function(x){
  sampledeck<-sample(x,15)
  deck1<-sampledeck[1:5]
  deck2<-sampledeck[6:10]
  deck3<-sampledeck[11:15]
  acettrue<-ifelse(
    length(which(deck1==1))==2|
    length(which(deck2==1))==2|
    length(which(deck3==1))==2, TRUE, FALSE)
  return(acettrue)
}

aces2<-replicate(aces(deck), n=10000)
table(aces2==TRUE)/10000
```

```
##
## FALSE TRUE
## 0.8771 0.1229
```

```
deck<-rep(1:13,4)
acetrue<-rep(NA,10000)
acetrue<-c()
for (i in 1:10000){
  sampledeck<-sample(deck,15)
  deck1<-sampledeck[1:5]
  deck2<-sampledeck[6:10]
  deck3<-sampledeck[11:15]
  acetrue[i]<-ifelse(
    length(which(deck1==1))==2|
    length(which(deck2==1))==2|
    length(which(deck3==1))==2, TRUE, FALSE)
}

table(acetrue==TRUE)/10000
```

```
##
## FALSE TRUE
## 0.8753 0.1247
```

- (b) As we can see, the probability that (n-1) letters will be placed in the correct envelope is 0 (In this case, we have 10 envelopes and 10 letters, the probability that 9 letters will be placed in the correct envelope is 0). Theoretically, without replacement, you can never have (n-1) letters put in correct envelopes and 1 is incorrect. If 1 is incorrect, there must be another one incorrect.

```
elmatch<-function(n){
  e<-sample(1:n,n,replace=F)
  l<-sample(1:n,n,replace=F)
  match<-e==l
  match
  totalmatch<-sum(match)
  return(totalmatch)
}

sim10<-replicate(elmatch(10), n=10000)
table(sim10)/sum(table(sim10))
```

```
## sim10
##      0      1      2      3      4      5      6      7      8
## 0.3632 0.3667 0.1865 0.0622 0.0172 0.0033 0.0006 0.0001 0.0002
```

- (c) The probabilities are calculated as below. We can compare the probability of 0 respectively. In this case, when n is 100, I have the largest q_n .

```
elmatch<-function(n){
  e<-sample(1:n,n,replace=F)
  l<-sample(1:n,n,replace=F)
  match<-e==l
```

```

match
totalmatch<-sum(match)
return(totalmatch)
}

sim10<-replicate(elmatch(10), n=10000)
table(sim10)/sum(table(sim10))

```

```

## sim10
##      0      1      2      3      4      5      6
## 0.3666 0.3627 0.1897 0.0640 0.0141 0.0026 0.0003

```

```

sim21<-replicate(elmatch(21), n=10000)
table(sim21)/sum(table(sim21))

```

```

## sim21
##      0      1      2      3      4      5      6
## 0.3659 0.3692 0.1822 0.0642 0.0143 0.0035 0.0007

```

```

sim53<-replicate(elmatch(53), n=10000)
table(sim53)/sum(table(sim53))

```

```

## sim53
##      0      1      2      3      4      5      6
## 0.3659 0.3672 0.1861 0.0622 0.0147 0.0036 0.0003

```

```

sim100<-replicate(elmatch(100), n=10000)
table(sim100)/sum(table(sim100))

```

```

## sim100
##      0      1      2      3      4      5      6      7
## 0.3669 0.3733 0.1769 0.0641 0.0157 0.0023 0.0007 0.0001

```

(d) The expected value is calculated as below. The value is very close to 1.

```

elmatch<-function(n){
match<-rep(NA,n)
e<-sample(c(1:n),replace=F)
l<-sample(c(1:n),replace=F)
for (i in 1:n){
match[i]<-ifelse(e[i]==l[i], TRUE, FALSE)
}
totalmatch<-sum(match==TRUE)
return(totalmatch)
}

sim10<-replicate(elmatch(10), n=10000)
table(sim10)/sum(table(sim10))

```

```

## sim10
##      0      1      2      3      4      5      6
## 0.3657 0.3622 0.1890 0.0615 0.0169 0.0039 0.0008

```

```
sum(sim10)/10000
```

```
## [1] 1.0166
```

```
mean(sim10)
```

```
## [1] 1.0166
```

```
sim20<-replicate(elmatch(20), n=10000)
```

```
sum(sim20)/10000
```

```
## [1] 0.998
```

```
sim50<-replicate(elmatch(50), n=10000)
```

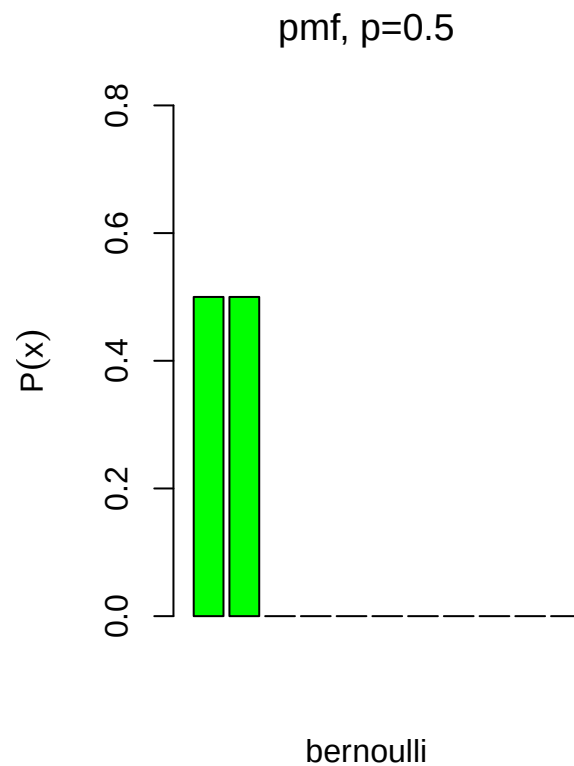
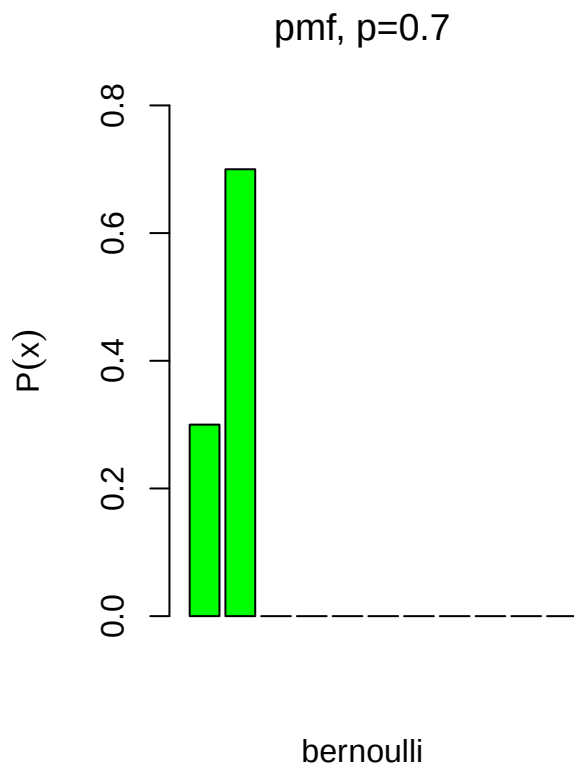
```
sum(sim50)/10000
```

```
## [1] 0.9889
```

8) *Answer to Problem 8*

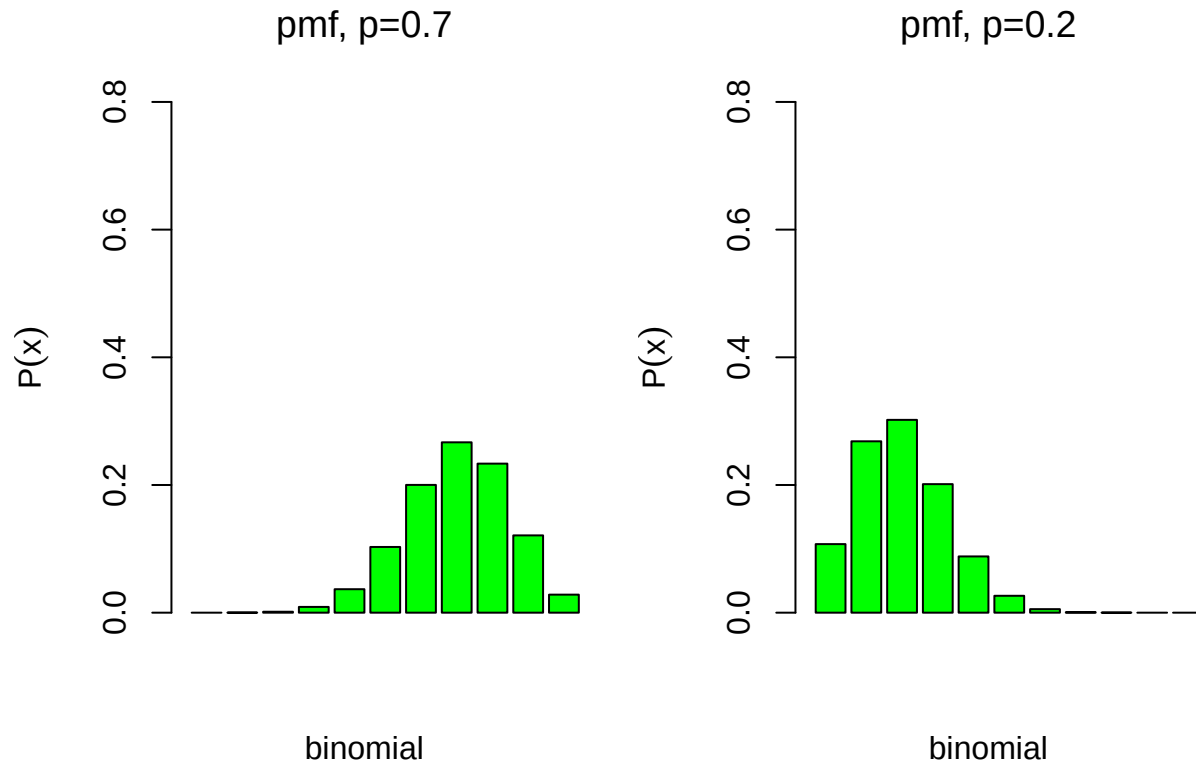
a) bernoulli

```
par(mfrow=c(1,2))
ber1=dbinom(0:10, 1, .7)
barplot(ber1,xlim=c(0,12),ylim=c(0,.8),col="green",xlab="bernoulli",ylab=expression(P(x)))
title(main = "pmf, p=0.7", font.main = 1)
ber2=dbinom(0:10, 1, .5)
barplot(ber2,xlim=c(0,12),ylim=c(0,.8),col="green",xlab="bernoulli",ylab=expression(P(x)))
title(main = "pmf, p=0.5", font.main = 1)
```



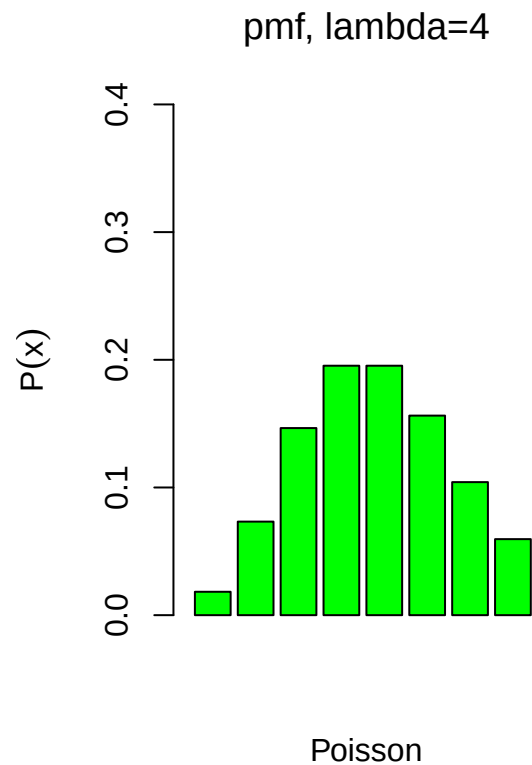
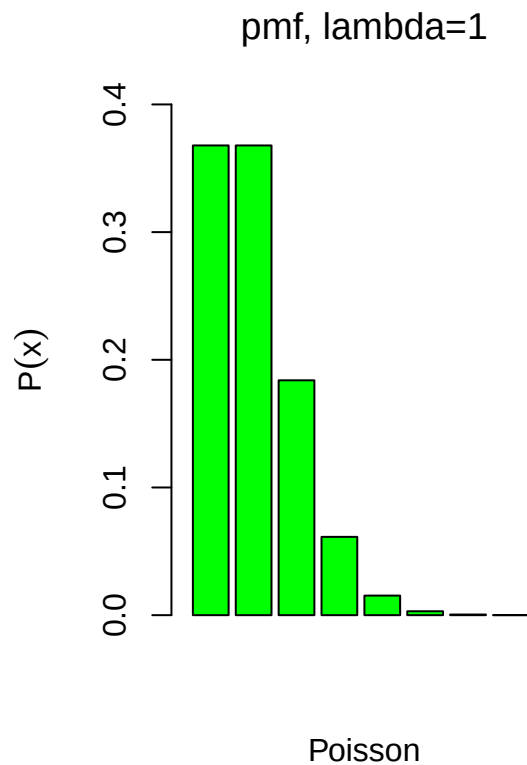
b) binomial

```
par(mfrow=c(1,2))
bin1=dbinom(0:10, 10,.7)
barplot(bin1,xlim=c(0,12),ylim=c(0,.8),col="green",xlab="binomial",ylab=expression(P(x)))
title(main = "pmf, p=0.7", font.main = 1)
bin2=dbinom(0:10, 10,.2)
barplot(bin2,xlim=c(0,12),ylim=c(0,.8),col="green",xlab="binomial",ylab=expression(P(x)))
title(main = "pmf, p=0.2", font.main = 1)
```



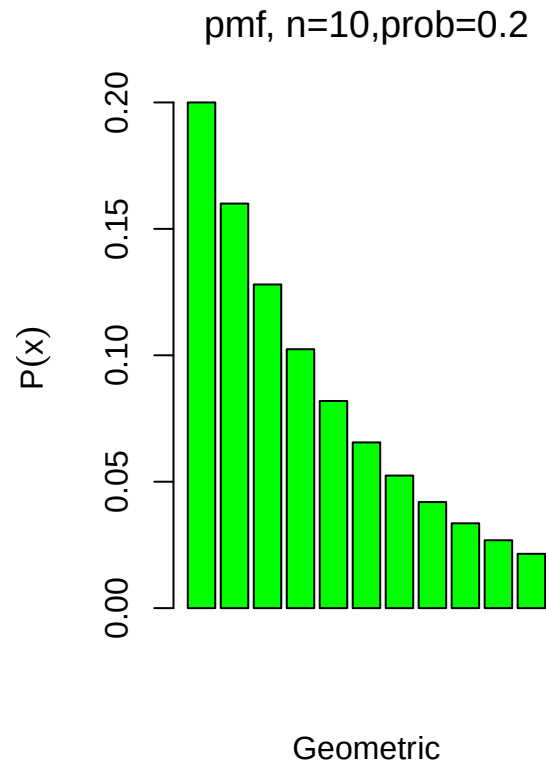
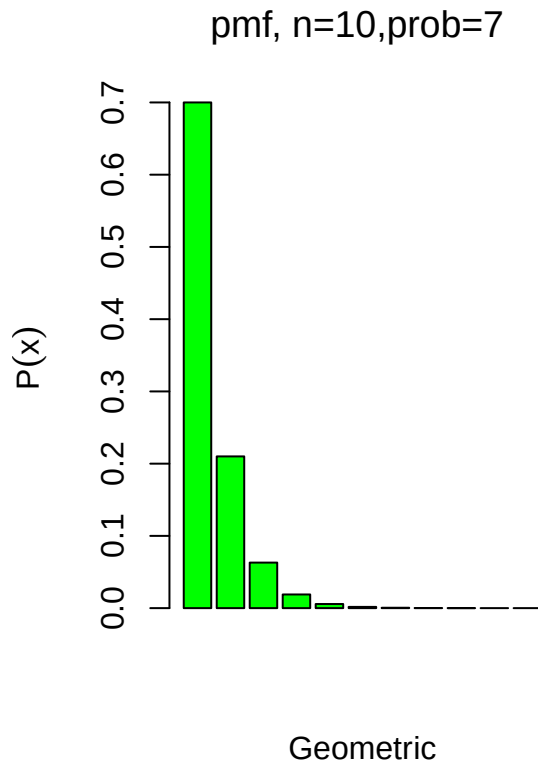
c) Poisson

```
par(mfrow=c(1,2))
po1=dpois(0:7, lambda=1)
barplot(po1,xlim=c(0,10),ylim=c(0,.4),col="green",xlab="Poisson",ylab=expression(P(x)))
title(main = "pmf, lambda=1", font.main = 1)
po2=dpois(0:7, lambda=4)
barplot(po2,xlim=c(0,10),ylim=c(0,.4),col="green",xlab="Poisson",ylab=expression(P(x)))
title(main = "pmf, lambda=4", font.main = 1)
```

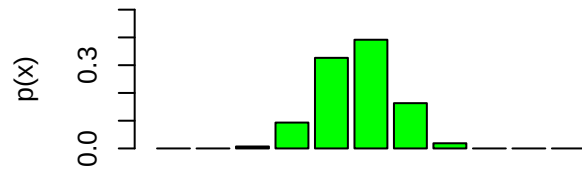
d) geometric

```
par(mfrow=c(1,2))
geom1 <- dgeom(0:10, 0.7, log=FALSE)
barplot(geom1, ,col="green",xlab="Geometric",ylab=expression(P(x)))
title(main = "pmf, n=10,prob=7 ", font.main = 1)
geom2 <- dgeom(0:10, 0.2, log=FALSE)
barplot(geom2,col="green", xlab="Geometric",ylab=expression(P(x)))
title(main = "pmf, n=10,prob=0.2", font.main = 1)
```

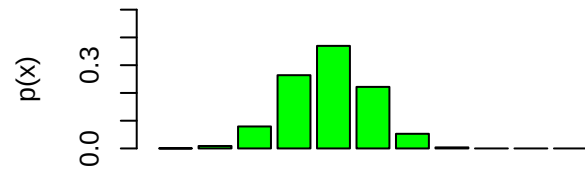


e) hypergeometric

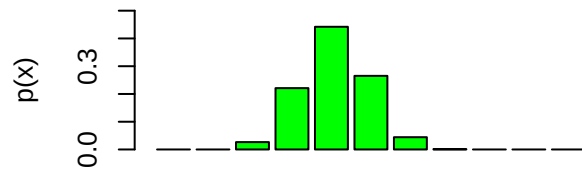
```
par(mfrow=c(2,2))
p=dhyper(0:10, 10, 5, 7) #change in n
barplot(p,xlim=c(0,12),ylim=c(0,.5),col="green",xlab="m=10,n=5,k=7",ylab="p(x)")
p=dhyper(0:10, 10, 8, 7) #change in n
barplot(p,xlim=c(0,12),ylim=c(0,.5),col="green",xlab="m=10,n=8,k=7",ylab="p(x)")
p=dhyper(0:10, 7, 5, 7) #change in m
barplot(p,xlim=c(0,12),ylim=c(0,.5),col="green",xlab="m=7,n=5,k=7",ylab="p(x)")
p=dhyper(0:10, 10, 5, 8) #change in m
barplot(p,xlim=c(0,12),ylim=c(0,.5),col="green",xlab="m=10,n=5,k=8",ylab="p(x)")
```



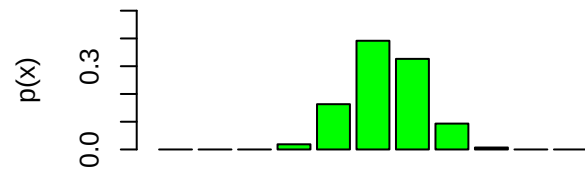
m=10,n=5,k=7



m=10,n=8,k=7



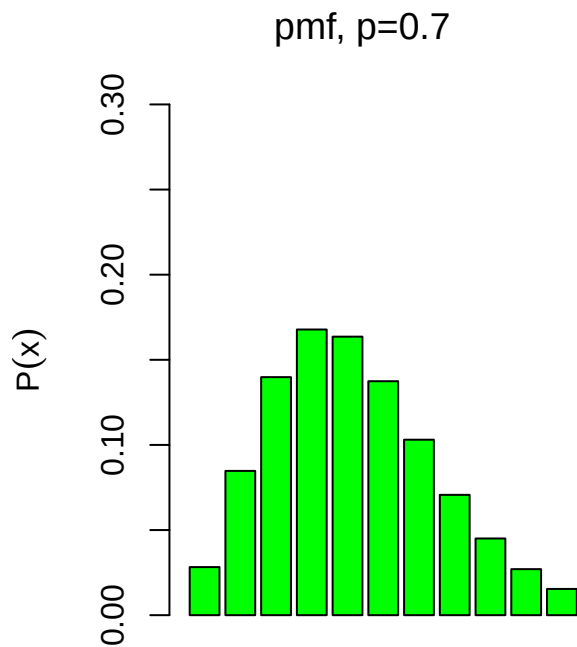
m=7,n=5,k=7



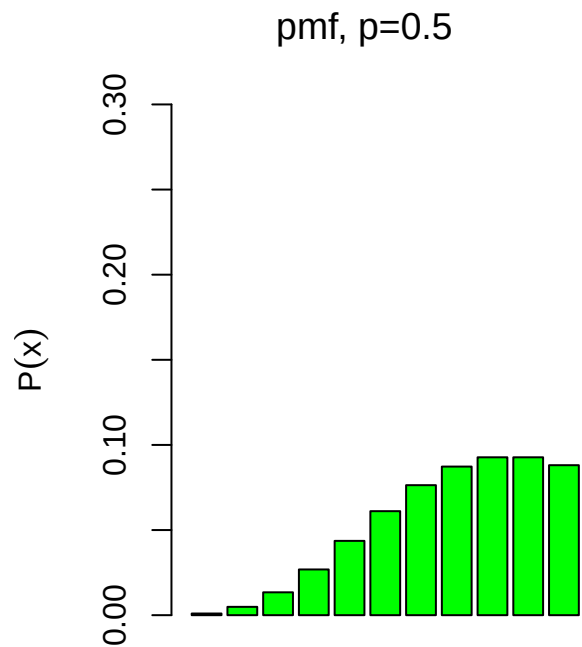
m=10,n=5,k=8

f) negative binomial

```
par(mfrow=c(1,2))
nbin1=dnbinom(0:10, 10,.7)
barplot(nbin1,xlim=c(0,12),ylim=c(0,.3),col="green",xlab="negative binomial",ylab=expression(P(x)))
title(main = "pmf, p=0.7", font.main = 1)
nbin2=dnbinom(0:10, 10,.5)
barplot(nbin2,xlim=c(0,12),ylim=c(0,.3),col="green",xlab="negative binomial",ylab=expression(P(x)))
title(main = "pmf, p=0.5", font.main = 1)
```



negative binomial

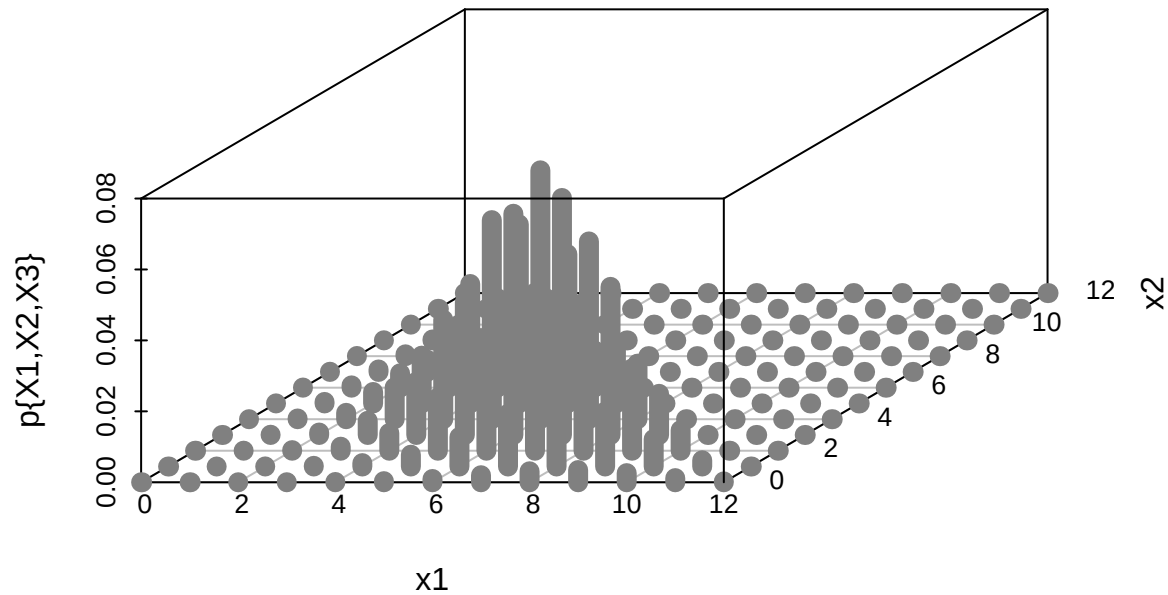


negative binomial

g)multinomial(trinomial)

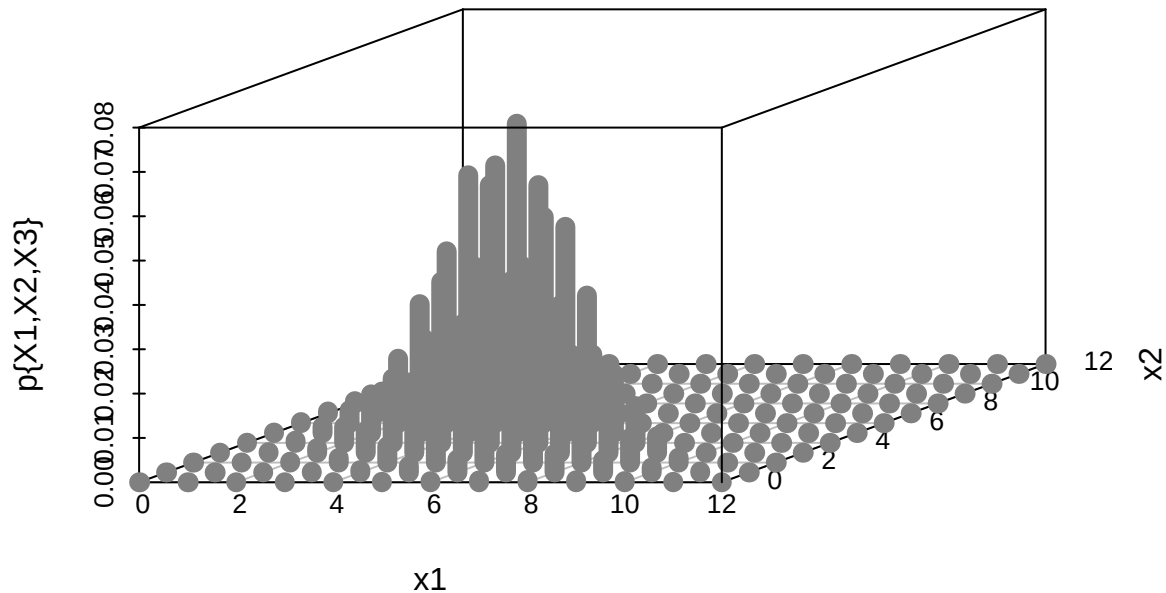
```
par(mfrow=c(1,1))
R = expand.grid(0:12, 0:12)
R[, 3] = 12 - R[, 1] - R[, 2]
names(R) = c("x1", "x2", "x3")
for (i in 1:nrow(R)) {
  v = R[i, 1:3]
  if (sum(v) == 12 & all(v >= 0)) {
    R$p[i] = dmultinom(v, prob = c(0.5, 0.3, 0.2))
  } else R$p[i] = 0
}
scatterplot3d(x = R$x1, y = R$x2, z = R$p, type = "h",
  lwd = 10, pch = " ", xlab = "x1", ylab = "x2",
  zlab = "p{X1,X2,X3}",
  color = grey(1/2))
title("Multinomial pmf, p(0.5,0.3,0.2)")
```

Multinomial pmf, $p(0.5,0.3,0.2)$



```
par(mfrow=c(1,1))
R = expand.grid(0:12, 0:12)
R[, 3] = 12 - R[, 1] - R[, 2]
names(R) = c("x1", "x2", "x3")
for (i in 1:nrow(R)) {
  v = R[i, 1:3]
  if (sum(v) == 12 & all(v >= 0)) {
    R$p[i] = dmultinom(v, prob = c(0.4, 0.4, 0.2))
  } else R$p[i] = 0
}
scatterplot3d(x = R$x1, y = R$x2, z = R$p, type = "h",
  lwd = 10, pch = " ", xlab = "x1", ylab = "x2",
  zlab = "p{X1,X2,X3}",
  color = grey(1/2))
title("Multinomial pmf, p(0.4,0.4,0.2)")
```

Multinomial pmf, $p(0.4,0.4,0.2)$



9) Answer to Problem 9

```
x <- c(89,37,30,28,2)
p <- c(.40,.20,.20,.15,.05)
chisqt1 <- chisq.test(x, p = p)
str(chisqt1)
```

```
## List of 9
## $ statistic: Named num 9.99
## .. attr(*, "names")= chr "X-squared"
## $ parameter: Named num 4
## .. attr(*, "names")= chr "df"
## $ p.value : num 0.0406
## $ method : chr "Chi-squared test for given probabilities"
## $ data.name: chr "x"
## $ observed : num [1:5] 89 37 30 28 2
## $ expected : num [1:5] 74.4 37.2 37.2 27.9 9.3
## $ residuals: num [1:5] 1.6926 -0.0328 -1.1805 0.0189 -2.3938
## $ stdres : num [1:5] 2.1852 -0.0367 -1.3198 0.0205 -2.456
## - attr(*, "class")= chr "htest"
```

```
?chisq.test
```

statistics: the value of the chi-squared test statistics.

parameter: the degree of freedom of the approximate chi-squared distribution of the test statistic. The degree of freedom here is 4.

p.value: the p-value of the test. It is 0.0406 here. It suggests that we can reject the null hypothesis at 0.05 level.

method: a character string indicating the type of test performed. The method here is “chi-squared test for given probabilities”

data.name: a character string giving the names of the data. The name of the data in this case is “x”

observed: the observed counts, here is a 5-element vector $c(89,37,30,28,2)$

expected: the expected counts under the null hypothesis, which is $c(74.4, 37.2, 37.2, 27.9, 9.3)$

residuals: the Pearson residuals, $(\text{observed}-\text{expected})/\sqrt{\text{expected}}$. The value greater than 2 or 3 indicates a lack of fit of the null hypothesis in the cell.

stdres: standardized residuals, $(\text{observed}-\text{expected})/\sqrt{V}$, where V is the residual cell variance. This is a measure of strength of the difference between observed and expected value. When you compare the cells, the standardized residual makes it easy to see which cells are contributing the most to the value, and which are contributing the least. If the cell's observed frequency is less than -2, the cell's observed frequency is less than the expected frequency. Greater than 2 and the observed frequency is greater than the expected frequency.